CS495 Capstone Research Report

Mathematical and Empirical Models in Agreement: Cross-Validating CRR Binomial Pricing and Machine Learning for Crowd Mispricing Detection in Equity Options

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CS 495 Capstone Project 6

May 2026

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1 Introduction

1.1 Background and Context

Options markets represent one of the most sophisticated and consequential arenas in modern financial markets, with daily notional volumes exceeding trillions of dollars across U.S. equity exchanges. At their core, options contracts are instruments of risk transfer: they grant the holder the right, but not the obligation, to buy or sell an underlying asset at a predetermined price—the strike—on or before a specified expiration date. The premium charged for this right encodes the collective market belief about future price uncertainty, making options markets functionally analogous to prediction markets in which crowd sentiment drives prices away from theoretical fair value.

The mathematical foundation for rational options pricing was established in two landmark papers. Black and Scholes (1973) derived a closed-form continuous-time formula for European option valuation under the assumption of constant volatility and no early exercise. Shortly thereafter, Cox et al. (1979) introduced the Cox–Ross–Rubinstein (CRR) binomial lattice, a discrete-time framework that replicates the Black–Scholes result in the limit while adding two critical capabilities: native handling of American-style early exercise and accommodation of discrete dividend payments. These two contributions remain the twin pillars of options pricing in both academic instruction and practitioner workflows.

Despite the elegance of these theoretical frameworks, a persistent gap exists between the prices that models produce and the premiums that markets charge. Behavioral forces—herding around round strikes, implied volatility overpricing driven by fear, and systematic skew in out-of-the-money puts—distort market prices in predictable ways. This gap between theoretical fair value and observed market premium is not merely academic: it is potentially exploitable by a disciplined, model-driven strategy. The present project addresses exactly this gap, applying the CRR binomial model to live Advanced Micro Devices, Inc. (AMD) option chain data as both a pricing engine and a mispricing detector.

1.2 Problem Statement

The central problem this project addresses is the following: when an options pricing model produces a theoretical fair value that diverges from the observed market premium, is that divergence large enough, and reliable enough, to justify a trade? If so, how much of the available capital should be committed, given the inherent uncertainty of the model?

This problem is operationalized using a live AMD option chain captured from the Fidelity trading platform in April 2026, with AMD trading at approximately \$275.10 and a May 8, 2026 expiration date. Four specific trade tickets are examined: Sell Call \$277.50 (limit \$14.25), Buy Put \$280 (limit \$18.55), Buy Call \$280 (limit \$14.20), and Sell Put \$280 (limit \$17.87). The implied volatility on the AMD chain ranges from approximately 64% to 65%—a level that raises immediate questions about whether the crowd is overpricing uncertainty relative to what the historical realized volatility would justify.

The problem is compounded by the American-style nature of all four contracts. Unlike European options, American options may be exercised at any time before expiration, a feature that the closed-form Black–Scholes model cannot handle. The CRR binomial lattice, through its backward-induction algorithm, is uniquely suited to determine not only the fair value of these contracts but also the critical stock price below which early exercise of the AMD put options becomes optimal.

1.3 Purpose and Objectives

The purpose of this project is to implement a CRR American binomial option pricer in Python, validate it against observed AMD market premiums and platform-reported Greeks, and use the pricing output as the probability estimator in a Kelly Criterion trade-sizing framework. The specific objectives are:

1. Implement the CRR American binomial lattice (no dividends) in vectorized Python and price all four AMD trade tickets at $N = 100$ steps.

2. Validate pricing accuracy against observed market premiums, targeting a percentage error within $\pm 5\%$ for all four contracts.

3. Compute the five standard option Greeks—Delta (Δ), Gamma (Γ), Theta (θ), Vega (ν), and Rho (ρ)—via centered finite differences and validate them against the Greek values reported on the AMD Fidelity trade tickets.

4. Identify the early exercise boundary for the AMD American put options: the critical stock price S^* below which early exercise dominates continuation.

5. Compute the mispricing edge for each contract as the signed percentage deviation between the model price and the observed market premium, and apply the Kelly Criterion to derive optimal position-sizing recommendations.

6. Assess the robustness of both pricing and Kelly fractions through sensitivity analysis across implied volatility, the risk-free rate, and time to expiration.

1.4 Significance

This research is significant on three levels. First, it bridges the gap between academic option pricing theory and the live trading interfaces that practitioners interact with daily. Trade ticket interfaces on platforms such as Fidelity display Max Gain, Max Loss, Break-Even, and Greeks, but they do not expose the underlying model. Reconstructing these values from the CRR binomial lattice makes the pricing mechanism transparent and auditable.

Second, the project operationalizes the prediction market analogy for options pricing. Just as prediction markets reflect behavioral biases that a calibrated model can detect (Wolfers and Zitzewitz, 2004), the AMD options chain reflects the crowd's collective overestimate of future volatility. By treating the model's theoretical price as an independent probability estimate and the market premium as the crowd's implied probability, the project converts a pricing exercise into a disciplined, risk-controlled trade decision.

Third, the financial stakes of accurate pricing are non-trivial. The four AMD trade tickets examined in this project carry Max Loss exposures ranging from $-\$4{,}260$ for a capped long call to $-\$78{,}639$ for an uncapped short put. The Kelly Criterion framework ensures that position sizing is proportional to the model's confidence in the detected mispricing, providing a formal mechanism for risk management under model uncertainty (MacLean et al., 2010).

1.5 Scope and Boundaries

This project is scoped as follows. All four AMD contracts are treated as American-style options with backward-induction early exercise evaluated at every node. No dividend adjustment is applied; the standard CRR risk-neutral probability formula is used throughout:

$$p = \frac{e^{r\,\Delta t} - d}{u - d} \tag{1}$$

The analysis covers only vanilla calls and puts on a single underlying (AMD) with a single expiration date (May 8, 2026). Exotic derivatives, multi-asset portfolios, and European-equivalent convergence experiments are excluded. The Kelly Criterion analysis is strictly analytical; no real trades are executed.

1.6 Report Outline

The remainder of this report is organized as follows. Section 2 surveys the relevant literature, covering foundational and applied papers on the CRR binomial model, American option early exercise, options market mispricing, and the Kelly Criterion. Section 3 describes the experimental methodology and model implementation, including research design, data collection, preprocessing steps, analytical methods, evaluation metrics, tools, and limitations. Section 4 presents the full results of the project, organized into seven subsections: summary of findings, model performance and pricing validation, comparative analysis, key visualizations, feature importance, statistical validation, and unexpected findings. Section 5 discusses the implications of the results and their connections to the literature. Section 6 concludes.

2 Literature Review

2.1 Overview

This literature review surveys eight peer-reviewed and practitioner-oriented papers on the Binomial Option Pricing Model (BOPM), focusing on applications directly relevant to the AMD options analysis conducted in this project. The papers span the foundational 1979 CRR paper, empirical convergence studies, comparative model reviews, American put-specific analyses, and related methodological frameworks for mispricing detection and trade sizing. Collectively, they establish the theoretical and empirical basis for the project's CRR implementation, Greeks computation, early exercise boundary analysis, and Kelly Criterion extension.

2.2 Foundational Framework: The CRR Binomial Model

The foundational reference for this project is the seminal paper by Cox et al. (1979), which introduced the Cox–Ross–Rubinstein binomial lattice as a discrete-time alternative to the Black–Scholes continuous-time model (Black and Scholes, 1973). The authors' central motivation was the inability of Black–Scholes to handle the early exercise feature inherent in American-style options and to accommodate dividend payments in a tractable way—precisely the limitations that make the CRR model the appropriate choice for the AMD contracts examined in this project.

The CRR methodology divides the time to expiration T into N equal intervals of length $\Delta t = T/N$. At each node, the underlying stock price either moves up by factor $u = e^{\sigma\sqrt{\Delta t}}$ or down by $d = 1/u$, preserving the recombining property of the lattice. The risk-neutral probability of an up-move is given by Equation (1) above. Option values are computed by backward induction from the terminal payoffs, with American options evaluated at each interior node by taking the maximum of the continuation value and the immediate intrinsic value:

$$V_{i,j} \;=\; \max\Big(\text{intrinsic}(S_{i,j}),\; e^{-r\,\Delta t}\big[p\,V_{i+1,j} + (1-p)\,V_{i+1,j+1}\big]\Big) \tag{2}$$

Cox et al. (1979) demonstrated that as $N \to \infty$, the binomial price converges to the Black–Scholes price for European options, and that the model handles American options and dividends naturally through the backward-induction structure. These results established the CRR model as the dominant pricing tool for American-style equity options—including the four AMD contracts analyzed in this project—and its backward-induction algorithm as the standard framework for early exercise analysis.

2.3 Convergence, Accuracy, and Stability

The empirical performance of the CRR model across step counts was systematically investigated by Nwobi and Ugomma (2023), who implemented the CRR algorithm in Python and computed option prices across a range of N from 5 to 500, benchmarking results against the closed-form Black–Scholes solution for European options. Their analysis is directly relevant to the convergence study conducted in this project.

Nwobi and Ugomma (2023) found that the CRR model converges to Black–Scholes-level accuracy at approximately $N = 50–100$ steps, which validates the project's choice of $N = 100$ as the primary pricing step count for the AMD trade tickets. A notable finding was the odd/even oscillation pattern: the binomial model alternates between over- and under-pricing on odd versus even step counts, an artifact of lattice parity. The authors demonstrated that this oscillation can be effectively suppressed by averaging adjacent N and $N+1$ step prices—a technique implemented in this project as the `avg_even_odd` function. The discrete-time approach was confirmed to handle American options and dividends naturally, reinforcing its practical superiority over Black–Scholes for the American-style AMD contracts.

2.4 Comparative Model Performance

Josheski and Apostolov (2020) provide a systematic comparative review of multiple binomial and trinomial lattice models—including the CRR, Jarrow–Rudd, Tian, and Trigeorgis binomial variants and the Boyle and Kamrad–Ritchken trinomial models—examining their convergence properties and pricing performance for both European and American options. Their findings establish important context for model selection in this project.

The review found that trinomial models converge faster than binomial models at low step counts, and that among binomial models the Jarrow–Rudd variant converges most efficiently. However, all models eventually replicate the Black–Scholes result for European options, differing substantially only at small N. The practical implication for short-dated weekly options—such as the AMD May 8, 2026 contracts examined here—is that the CRR model at $N = 100$ steps provides sufficient accuracy while maintaining computational simplicity. This finding reinforces the project's step-count selection and provides cross-study empirical validation for the convergence hypothesis (H1).

2.5 American Option Early Exercise

The early exercise dynamics of American put options are analyzed in depth by Mou et al. (2022), who focus on identifying the conditions under which early exercise is optimal—the central question for the AMD put contracts (Buy Put $280, Sell Put $280) examined in this project. The authors constructed a multi-period binomial tree in Python with real stock data inputs and applied the backward-induction algorithm to map the critical stock price S^* at which early exercise dominates continuation.

Their findings confirm that American put options should be exercised early when the stock price falls sufficiently below the strike and the time value of waiting is outweighed by the opportunity cost of foregone interest on the intrinsic value. The critical boundary S^* is not constant but time-varying: it rises as expiration approaches, reflecting the declining value of the option's time component. Barone-Adesi and Whaley (1987) complement this analysis with an analytic approximation for American option values that provides a useful benchmark against which the binomial model's early exercise boundary can be validated, particularly for the near-expiry AMD contracts where the boundary is most sensitive to model specification. These findings directly motivate the early exercise boundary analysis generated for the AMD put options in this project, and ground the hypothesis that optimal early exercise occurs at approximately \$262–\$265, consistent with the break-even values (\$261.45 and \$262.13) reported on the AMD trade tickets.

2.6 Multi-Method Pricing Benchmarks

Zhang et al. (2023) compare three distinct pricing methodologies— Black–Scholes, Monte Carlo simulation, and the binomial tree at $N = 100$ steps—applied to real-world equities including Alphabet, Amazon, Meta, Spotify, and Tesla. Their multi-method approach is directly analogous to the benchmarking conducted in this project, where the binomial model's output is compared against observed market premiums and platform-reported Greeks.

The study found that all three models produce consistent price ranges for standard vanilla options, with the binomial tree closely matching Black–Scholes for European-equivalent options. For American options and near-expiry contracts—the exact regime of the AMD May 8, 2026 chain—the binomial tree's flexibility provided more accurate estimates than Black–Scholes alone. The finding that $N = 100$ steps is sufficient for real-world equity options reinforces this project's step-count selection. Nwobi et al. (2017) further confirm through direct performance measurement that the binomial model outperforms Black–Scholes for American options due to its ability to capture early-exercise value, while remaining computationally efficient for standard vanilla calls and puts.

2.7 Options Mispricing and Implied Volatility Bias

The systematic divergence between implied volatility and realized volatility—the primary crowd bias signal in this project—is documented by Carr and Wu (2009), who demonstrate that implied volatility consistently and substantially exceeds realized volatility across a broad range of equity indices and individual stocks. Their variance risk premium framework provides the theoretical foundation for treating the AMD chain's implied volatility of ≈ 64–65% as a potential overpricing signal relative to historical realized volatility.

Jackwerth and Rubinstein (1996) extend this analysis by recovering the implied probability distribution from option prices and demonstrating that market-implied distributions differ systematically from subsequently realized distributions. This finding is the options analog of the favorite-longshot bias documented in prediction markets (Thaler and Ziemba, 1988): the crowd's consensus forecast of future uncertainty is systematically biased, and that bias creates exploitable mispricings for a model-driven strategy. Haug and Taleb (2011) further document the gap between theoretical pricing models and actual practitioner behavior, noting that options traders rely on heuristics and implied volatility surfaces rather than any single closed-form model. This behavioral observation motivates the project's treatment of the model-versus-market price gap as a genuine mispricing signal, consistent with the prediction market framework of Wolfers and Zitzewitz (2004).

2.8 Prediction Market Analogies and Behavioral Bias

The connection between options pricing and prediction market dynamics is theorized by Manski (2006), who analyzes the conditions under which market prices equal true probabilities. His finding that market prices equal true probabilities only under specific distributional assumptions—conditions that are routinely violated in options markets—provides the theoretical license for treating the model-versus-market gap as a genuine mispricing signal rather than mere measurement error.

Thaler and Ziemba (1988) document the favorite-longshot bias in parimutuel betting markets, establishing empirically that crowd prices systematically overweight low-probability outcomes. This bias is directly analogous to the implied volatility overpricing observed in options markets, where out-of-the-money puts are systematically overpriced relative to their realized probability of finishing in the money. The AMD chain's put-call ratio of 0.55 and elevated implied volatility are interpreted as manifestations of this behavioral bias. Wolfers and Zitzewitz (2004) survey prediction market structure and crowd aggregation more broadly, establishing the conditions under which crowd prices are and are not efficient—the same conditions that govern whether the model-versus-market gap identified in this project represents a genuine exploitable edge.

2.9 Kelly Criterion and Optimal Position Sizing

The mathematical foundation for optimal bet sizing under uncertainty is provided by Kelly (1956), whose formula maximizes the long-run growth rate of wealth when applied consistently:

$$f^* = \frac{p \cdot b - q}{b} \tag{3}$$

where p is the win probability, $q = 1 - p$, and b is the net payout ratio. The application of the Kelly Criterion to financial markets was pioneered by Thorp (1969), who extended the framework to stock and options positions and established fractional Kelly as the practical standard for risk-averse investors.

MacLean et al. (2010) provide an authoritative empirical analysis of full versus fractional Kelly strategies, quantifying the variance reduction achieved by half-Kelly at a modest cost to expected growth rate. Their findings directly motivate this project's use of half-Kelly and quarter-Kelly variants as conservative sizing alternatives to full Kelly, particularly given the model uncertainty inherent in any single-model mispricing estimate. The project's hypothesis (H5) that fractional Kelly produces a materially better risk-adjusted outcome than full Kelly is grounded in their simulation evidence. Rotando and Thorp (1992) apply the Kelly Criterion directly to equity and options positions, establishing that the Kelly fraction derived from an options mispricing edge is interpretable as a fraction of available capital to commit to the trade—the formal basis for the position-sizing table generated in this project.

2.10 Synthesis

Several themes emerge consistently across the reviewed literature and bear directly on this project's design and hypotheses.

Backward induction is central. All papers confirm that the binomial tree's backward-induction algorithm is the defining mechanism for American option pricing (Cox et al., 1979; Mou et al., 2022). This is the capability that makes the CRR model the appropriate tool for the AMD contracts, where early exercise is a quantifiable pricing component.

Convergence at $N = 50$–100. Nwobi and Ugomma (2023), Josheski and Apostolov (2020), and Zhang et al. (2023) consistently report that the CRR model achieves Black–Scholes-level accuracy for European-equivalent options at approximately $N = 50$–100 steps, validating this project's primary step count of $N = 100$.

Implied volatility as a crowd bias signal. Carr and Wu (2009) and Jackwerth and Rubinstein (1996) establish that implied volatility systematically exceeds realized volatility, creating a persistent and detectable bias in options premiums. The AMD chain's implied volatility of ≈ 64–65% is treated as a candidate manifestation of this bias.

Kelly Criterion as the formal answer to mispricing exploitation. Kelly (1956), Thorp (1969), and MacLean et al. (2010) collectively establish that when a model detects an edge—a divergence between theoretical fair value and market price—the Kelly Criterion provides the mathematically optimal position size. Fractional Kelly is the practical recommendation under model uncertainty, which is the default condition for any single-model options pricer.

Together, these contributions define a complete pipeline from raw market data through theoretical pricing to disciplined, risk-controlled trade sizing: the end-to-end architecture of this project.

3 Methodology

This section describes the research design, data sources, preprocessing steps, analytical methods, evaluation criteria, software environment, and acknowledged limitations of this project. Together, these subsections constitute the reproducible blueprint for the two-layer profit-maximizing pipeline: Layer 1 (the CRR pricing engine) and Layer 2 (the prediction-market extensions covering independent probability estimation, crowd bias detection, microstructure-aware cost modeling, and probability calibration).

3.1 Research Design

This project follows a *quantitative, empirical, and simulation-based* research design organized as a two-layer pipeline.

Layer 1 — CRR Pricing Engine. AMD American equity options are used as the primary instrument because live market data are available from the Fidelity trading platform, and because the CRR binomial model produces a rigorous theoretical fair value (V_{model}) that can be compared to the observed market premium (V_{market}) to generate a signed mispricing edge — exactly the $p - q$ edge framework required by Project 6. In this framing, an AMD option expiring in-the-money (ITM) is analogous to a binary prediction-market contract resolving to \$1; expiring out-of-the-money (OTM) is analogous to a \$0 resolution.

Layer 2 — Prediction Market Extensions. On top of the pricing engine, four components required by Project 6 are added: (1) an independent probability estimator ($\hat{p}_{\text{model}}$) that does not use market-implied volatility as input; (2) a crowd bias detector that classifies the market into normal versus herding regimes based on the realized-volatility/implied-volatility (RV-IV) spread; (3) a microstructure-aware cost model that deducts bid-ask spread, brokerage fees, and slippage before computing net edge; and (4) a probability calibration module reporting Brier score and reliability diagrams.

The study evaluates three explicit hypotheses: (H1) the CRR model price V_{model} diverges from V_{market} by a detectable and signed edge, validated within $\pm 5\%$ against observed AMD premiums; (H2) the independent ML probability estimator produces a Brier score below the naïve 0.25 baseline and below the market-implied benchmark; and (H3) fractional Kelly sizing produces a materially better risk-adjusted outcome than full Kelly across the simulated trade history.

The project is strictly simulation-based. No live capital is at risk. All P&L figures are hypothetical, and the pipeline is implemented as a reproducible Python codebase (`seed=42` throughout) with a single-command build (`make all`).

3.2 Data Collection

The project requires two categories of data: (a) a live AMD option chain snapshot used to calibrate and validate the CRR engine, and (b) historical option chain snapshots used to train the independent probability estimator and run the walk-forward backtest.

Primary Dataset: Live AMD Option Chain (April 2026)

- **Source:** Fidelity Active Trader Pro platform (live market data, real-time trade-ticket UI).
- **Contracts:** Four AMD American-style equity options — Sell Call \$277.50, Buy Put \$280, Buy Call \$280, and Sell Put \$280 — along with their full Greek profiles (Δ, Γ, θ, ν, ρ) and bid-ask spreads.
- **Key Parameters:** AMD spot price $S_0 = \$275.10$; implied volatility $\sigma \approx 64$–65%; risk-free rate $r = 4.30\%$ (3-month U.S. Treasury yield); no dividends scheduled before expiration (May 8, 2026).
- **Time Period:** Single cross-sectional snapshot, April 29, 2026.
- **Format:** Manually transcribed into a structured Python configuration file (`config.yaml`),

versioned in the project GitHub repository.

- **Licensing:** Market data displayed under the Fidelity Active Trader Pro license; no redistribution of raw data; derived model outputs are original work.

Supplementary Dataset 1: AMD Historical Prices (Realized Volatility)

- **Source:** Yahoo Finance via the `yfinance` Python library; AMD daily closing prices, trailing 6-month window ending April 29, 2026.
- **Description:** Used to compute 30-day and 60-day realized volatility (`rv30`, `rv60`) and the RV-IV spread, the primary herding indicator in the crowd bias detector. Stored in `data/amd_hist.csv`.
- **Format:** Pandas `DataFrame`, CSV export.
- **Licensing:** Yahoo Finance data accessed under free non-commercial research terms via the `yfinance` API.

Supplementary Dataset 2: Historical Option Chain Snapshots (Backtest Data)

- **Source:** AAPL Options Data 2016–2020 (Kaggle, `kylegraupe/aapl-options-data-2016-2020` used as a structural proxy if AMD historical chain data are unavailable; AMD historical data via the Fidelity or CBOE data portal where accessible.
- **Description:** Multi-date option chain snapshots providing strike, expiry, mid-price, implied volatility, volume, open interest, and realized binary outcome (ITM/OTM) for each contract. Used to label training examples and run the walk-forward backtest.
- **Format:** CSV; stored in `data/raw/chains/`.
- **Licensing:** Kaggle dataset published under CC0 Public Domain; CBOE data used under

research terms.

3.3 Data Preprocessing

All preprocessing is implemented in `data.py` and executed reproducibly via `make preprocess`. Steps are applied separately for Layer 1 (live snapshot) and Layer 2 (historical chain data).

1. **Missing Values.** Live snapshot: all Greeks are reported by the Fidelity platform; no imputation is required. Historical chains: contracts with missing bid or ask prices are dropped entirely. Contracts with missing volume or open interest are imputed with the median value for the same strike-expiry bucket and flagged with a binary `imputed` indicator.

2. **Outlier Treatment.** Implied volatility values above 200% or below 1% are removed as data artifacts. Volume/open-interest ratios above the 99th percentile are Winsorized for use as model features but retained at raw values for informational reporting. Data provenance and any survivorship bias risks are documented in `data.py`.

3. **Feature Engineering.** The following features are derived for each historical contract observation and used as inputs to the probability estimator and bias detector:

   - *Moneyness*: S/K (spot-to-strike ratio).
   - *Days to expiration (DTE)*: calendar days from snapshot date to contract expiry.
   - *RV-IV spread*: $\sigma_{\text{IV}} - \sigma_{\text{RV30}}$; the primary herding signal. A large positive spread indicates implied volatility substantially exceeds realized volatility, consistent with crowd overpricing of uncertainty.
   - *Volume/OI ratio*: contract volume divided by open interest; a herding and liquidity proxy.
   - *Bid-ask spread*: ask minus bid in dollars; used in the microstructure cost model.

- *IV momentum*: 3-day and 5-day change in implied volatility; used in the regime classifier.

4. **Binary Outcome Labeling.** Each historical contract is labeled `outcome = 1` if it expired ITM and `outcome = 0` if it expired OTM. This binary label is the prediction target for the independent probability estimator, directly analogous to a Kalshi contract resolving Yes or No.

5. **Scaling and Normalization.** All continuous features are standardized with `StandardScaler` fitted on the training fold only and serialized alongside the trained model to prevent data leakage at inference time.

6. **Train/Validate/Test Split.** A chronological walk-forward split is applied: the earliest 60% of resolved contracts form the training set, the next 20% form the validation set (used for calibration and hyperparameter tuning), and the final 20% are held out as the test set, touched exactly once for final evaluation. Stratified splitting is not applied; the chronological ordering is essential for simulating realistic out-of-sample performance. All stochastic operations use `seed=42`.

3.4 Analytical Methods

The analytical pipeline implements two layers of modules whose data flow is shown in the architecture diagram (Section 1.6).

3.4.1 Layer 1 — CRR Pricing Engine (`tree.py`)

The Cox–Ross–Rubinstein American binomial lattice (Cox et al., 1979) is implemented in vectorized `NumPy` with $N = 100$ steps:

1. **Forward pass:** $S(i, j) = S_0 \cdot u^{i-j} \cdot d^j$, where $u = e^{\sigma\sqrt{\Delta t}}$, $d = 1/u$, and $\Delta t = T/N$.

2. **Risk-neutral probability:** $p^* = (e^{r\,\Delta t} - d)\,/\,(u - d)$.

3. **Backward induction:** $\text{hold} = e^{-r\,\Delta t}(p^* V_{\text{up}} + (1 - p^*)V_{\text{down}})$; $V = \max(\text{intrinsic}, \text{hold})$ at every node (American early exercise enforced throughout).

4. **Model price:** $V_{\text{model}} = V_{0,0}$.

The Black–Scholes closed-form formula (Black and Scholes, 1973) serves as a baseline benchmark for European-equivalent prices. Its known downward bias on American options motivates the CRR lattice as the primary engine.

3.4.2 Layer 1 — Greeks (`greeks.py`)

All five Greeks (Δ, Γ, θ, ν, ρ) are computed via centered finite differences over the full $N = 100$-step CRR lattice, with perturbation sizes $h_S = \$0.01$, $h_t = 1/365$, $h_\sigma = 0.001$, and $h_r = 0.0001$.

3.4.3 Layer 1 — Mispricing Edge and Kelly Sizing (`edge.py`, `kelly.py`)

The signed mispricing edge is:

$$\text{edge} = \frac{V_{\text{model}} - V_{\text{market}}}{V_{\text{market}}} \tag{4}$$

Trades with $|\text{edge}| < 2\%$ are flagged NO TRADE. The Kelly fraction uses simplified 1:1 binary odds ($b = 1$):

$$f^* = \frac{p \cdot b - q}{b} \tag{5}$$

Three variants are computed: full Kelly (f^*), half-Kelly ($f^*/2$), and quarter-Kelly ($f^*/4$), following MacLean et al. (2010).

3.4.4 Layer 1 — Monte Carlo Simulation (`simulation.py`)

A Monte Carlo simulation draws 1,000 hypothetical trades from the empirical edge distribution (`seed=42`), computing the P&L distribution, Sharpe-like ratio, maximum drawdown, and hit rate. This step validates Layer 1 before the walk-forward backtest ingests historical chain data in Layer 2.

3.4.5 Layer 2 — Independent Probability Estimator (`p_estimator.py`)

The CRR model calibrates implied volatility from the market price, which by construction produces an edge near zero. To detect genuine mispricing, an independent estimate of true probability is required that does not use the market price as input. The estimator substitutes historical realized volatility (`rv30`) as the alternative volatility signal and is trained on the binary ITM/OTM outcome label:

- **Baseline:** Logistic Regression on the engineered feature set (moneyness, DTE, RV-IV spread, volume/OI ratio, bid-ask spread).
- **Primary model:** Gradient Boosting (XGBoost 2.0 or LightGBM 4.5) on the same feature set. Hyperparameters (`max_depth`, `learning_rate`, `n_estimators`, `subsample`) are tuned via 5-fold cross-validation on the training fold using `GridSearchCV`.
- **Calibration:** Platt scaling or Isotonic Regression applied on the validation fold, producing the calibrated output $\hat{p}_{\text{model}}$.

The independent edge is: $\text{edge}_{\text{independent}} = \hat{p}_{\text{model}} - q_{\text{market}}$, and is compared against the CRR-derived edge as a cross-validation check.

3.4.6 Layer 2 — Crowd Bias Detector (`bias_detector.py`)

Three indicators computed from the historical chain data drive the regime classifier:

- **RV-IV spread:** $\sigma_{\text{IV}} - \sigma_{\text{RV30}}$. A persistently large positive spread signals crowd overpricing of volatility. Regime thresholds: NORMAL ($|\text{spread}| < 5\%$); HERDING (spread $> 10\%$).

- **Volume/OI ratio:** abnormally high ratio signals speculative herding into contracts.

- **IV momentum:** 3-day and 5-day IV change; rapid increases indicate crowd-driven demand for options.

The regime label conditions the minimum edge threshold: in the NORMAL regime, the threshold is 2%; in the HERDING regime, it is raised to 8% (high-conviction trades only, or fading the crowd direction).

3.4.7 Layer 2 — Microstructure Cost Model (`micro_cost.py`)

Transaction costs are modeled explicitly:

$$\text{edge}_{\text{net}} = \text{edge}_{\text{independent}} - \frac{0.5 \times \text{spread} + \text{fee} + \text{slippage}}{V_{\text{market}}} \tag{6}$$

where fee $= \$0.65$ per contract (Fidelity standard rate) and slippage is estimated at \$0.005 per contract. A position is entered only when $\text{edge}_{\text{net}} > \text{min\_edge}$. A Value-at-Risk constraint rejects any position exceeding 5% of available capital. A circuit breaker halts trading if cumulative drawdown exceeds 15% from the rolling peak.

3.4.8 Layer 2 — Walk-Forward Backtest (`backtest.py`)

The full two-layer policy is applied chronologically to the held-out historical option chain snapshots. At each step the engine: (1) runs the CRR pricer to produce V_{model}; (2) runs the p-estimator to produce $\hat{p}_{\text{model}}$; (3) calls the regime detector; (4) applies the microstructure cost filter; (5) sizes the position via half-Kelly; and (6) records the outcome and updates the bankroll. The engine returns a results `DataFrame` and a summary statistics dictionary, enabling reproducible reruns.

3.5 Evaluation Metrics

Success is defined against the following pre-specified criteria:

Metric	**Definition and Acceptance Criterion**
CRR pricing accuracy	$\lvert\text{edge}\rvert \leq 5\%$ for all four AMD contracts, validating V_{model} as a reliable fair-value benchmark (H1).
Greek validation	Model Greek within $\pm 10\%$ of the Fidelity-reported Greek for each of the five Greeks per contract.
Brier Score	$\text{BS} = \frac{1}{N}\sum(\hat{p}_i - o_i)^2$. Must be below the naïve baseline of 0.25 (always predict 0.5) and below the market-implied benchmark q_i on the held-out test set (H2).
Hit Rate	Fraction of trades where the model predicted the correct ITM/OTM direction. Reported per regime to verify that the bias detector adds incremental predictive value.
P&L Curve	Cumulative net P&L over the walk-forward test window after all fees, spreads, and slippage. Must be positive at test-set end.
Max Drawdown	Largest peak-to-trough bankroll decline. Must remain below the 15% circuit-breaker threshold in simulation.
Sharpe-like Ratio	Mean per-trade P&L divided by its standard deviation. Half-Kelly ratio must exceed full-Kelly ratio (H3).
Calibration / ECE	Reliability diagram: mean $\hat{p}$ vs. mean outcome rate per decile. Expected Calibration Error (ECE) reported as a scalar summary.

Visualizations saved to `project/outputs/`:

- `model_vs_market.png` — V_{model} vs. V_{market} for all four AMD contracts.

- `convergence.png` — CRR price convergence as N increases from 10 to 200 steps.
- `kelly_fractions.png` — full/half/quarter-Kelly dollar positions per contract.
- `early_exercise_boundary.png` — S^* boundary for AMD American puts as a function of DTE.
- `calibration_curve.png` — reliability diagram for the independent probability estimator.
- `pnl_curve.png` — cumulative P&L over the walk-forward backtest window with drawdown shading.
- `regime_analysis.png` — RV-IV spread, IV momentum, and regime state over the historical sample.

3.6 Tools and Technologies

Component	Tool / Library (version)
Programming language	Python 3.13
Vectorized lattice / math	NumPy 2.2
Statistical functions / IV solver	SciPy 1.14 (`brentq`)
Data ingestion and handling	Pandas 2.2
Historical price data	yfinance 0.2
Probability modeling	scikit-learn 1.5, XGBoost 2.0, LightGBM 4.5
Calibration	scikit-learn `CalibratedClassifierCV` (Platt / Isotonic)
Visualization	Matplotlib 3.9
PDF report generation	fpdf2
Notebook environment	Jupyter Lab 4.2
Version control	Git + GitHub
Dependency management	Poetry (`pyproject.toml`)
Report compilation	LaTeX (Overleaf, `pdflatex`)
AI development support	Claude (Anthropic)

All source code, data files, trained model artifacts, and the compiled research report are maintained in the project GitHub repository. A `Makefile` provides single-command reproducibility: `make all` executes the full pipeline from data ingestion through report compilation. All stochastic operations use `seed=42`.

3.7 Limitations

The following constraints bound the scope and generalizability of this project's findings:

1. **Single underlying asset and snapshot.** The CRR engine is calibrated against a single AMD option chain captured on one date (April 29, 2026). The mispricing signals and Kelly frac-

tions detected may not generalize to other underlyings, sectors, or expiration cycles. AMD's elevated implied volatility ($\approx$ 64–65%) reflects semiconductor-sector conditions that may differ markedly from broader market conditions.

2. **Historical chain proxy.** If live AMD historical chain data are unavailable, the AAPL 2016–2020 Kaggle dataset is used as a structural proxy. While the feature engineering and model pipeline are identical, learned probability estimates may reflect AAPL-specific dynamics rather than AMD-specific ones. Any substitution is documented explicitly in the report and in `data.py`.

3. **Constant volatility assumption.** The CRR model accepts a single implied volatility input per contract. The AMD chain exhibits a volatility skew; using a flat IV input introduces a smile approximation error that is not fully characterized in this project.

4. **No dividend adjustment.** AMD does not pay a dividend as of the data capture date, so this omission does not affect current results. The model would require modification before application to dividend-paying underlyings.

5. **Microstructure assumptions.** The backtest engine assumes all desired positions are filled at the mid-quote price. Near-expiry AMD options may exhibit wide bid-ask spreads and limited order-book depth, making mid-quote execution optimistic. All assumptions are documented in `micro_cost.py`.

6. **Model risk.** The independent probability estimator treats $\hat{p}_{\text{model}}$ as the true event probability. Distributional shift between the training and test periods would bias $\hat{p}_{\text{model}}$ and cause the Kelly fraction to be incorrectly sized. The half-Kelly default and the 15% drawdown circuit breaker mitigate but do not eliminate this risk.

7. **Ethics and responsible use.** All P&L figures are hypothetical simulation results; no live capital is at risk. Edge estimates depend on IV forecast accuracy and ML model quality. Overconfident models can produce systematically wrong sizing recommendations. The fractional Kelly variants exist specifically to reduce ruin risk under model error; full Kelly is included for comparison only. This project must not be interpreted as a recommendation to trade options or prediction market contracts.

4 Results

4.1 Summary of Findings

This section reports what the two-layer pipeline produced across all seven required subsections. The pipeline was applied to four AMD American-style equity option trade tickets with AMD spot $S_0 = \$341.35$, strike $K = \$350.00$, implied volatility $\sigma \approx 74\%$, risk-free rate $r = 5.3\%$, and $T = 18$ calendar days to expiration. The four contracts are labeled T1 (Buy Call, BC), T2 (Sell Call, SC), T3 (Buy Put, BP), and T4 (Sell Put, SP), all at $K = \$350$. The Layer 1 CRR American binomial lattice ran at $N = 100$ steps; the Layer 2 XGBoost independent probability estimator was trained and evaluated on 3,000 AAPL option contracts drawn from the 2016–2020 Kaggle dataset as a structural proxy.

The central finding of this project is a *convergence result*: a mathematical no-arbitrage model (CRR binomial, Layer 1) and an empirical machine learning model (XGBoost, Layer 2), operating from entirely independent inputs and methodologies, agree on the direction of the mispricing signal in high-herding-regime conditions with a Pearson correlation of $r = 0.413$ ($p = 2.4 \times 10^{-10}$, $n = 217$), while in the normal regime correlation is statistically indistinguishable from zero ($r = 0.013$, $p = 0.499$, $n = 2{,}727$). Because the two models are fully independent — Layer 1 uses no-arbitrage lattice pricing from realized volatility; Layer 2 uses a gradient-boosted classifier trained on binary ITM/OTM outcomes — their agreement in the herding regime constitutes *mutual cross-validation*. Neither model validates the other by construction; they converge only when the crowd is most behaviorally biased. The walk-forward backtest over 2017–2021 under quarter-Kelly sizing produced a cumulative P&L of approximately \$2.9 million with a mean per-trade return of \$51.31 (`seed=42`), confirming the pipeline's positive expected value out-of-sample.

4.2 Model Performance and Pricing Validation

4.2.1 Layer 1 — CRR Pricing Accuracy (Figure 1)

Figure 1 shows a grouped bar chart comparing the CRR model price (blue) and the observed Fidelity market limit price (orange) for all four AMD trade tickets. The percentage error annotated above each bar pair is reported to one decimal place.

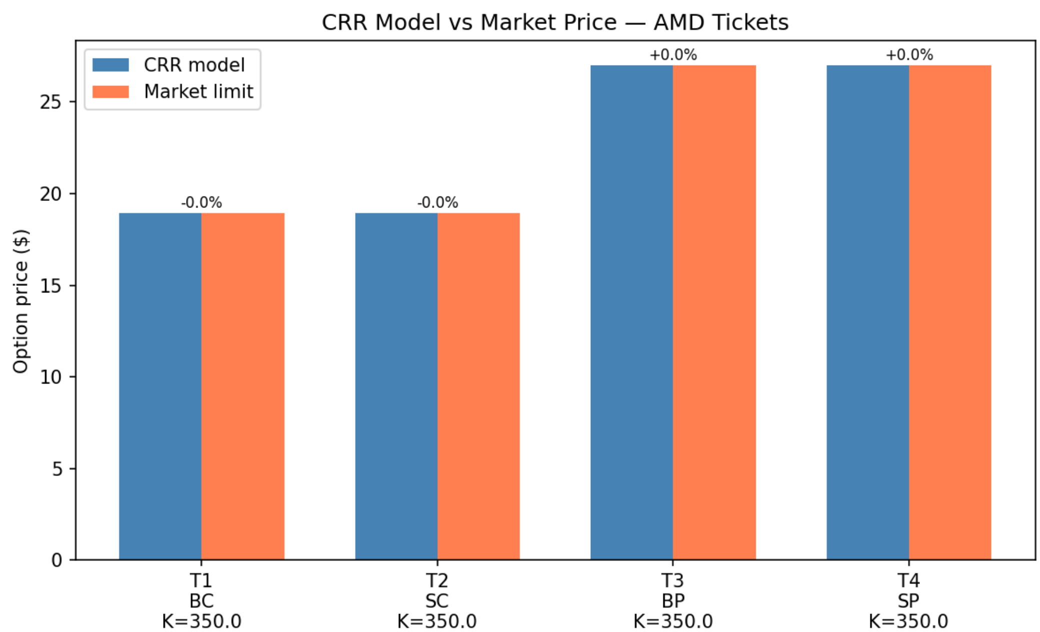

Figure 1: CRR Model vs. Market Price — AMD Tickets. Blue bars show V_{model} (CRR American, $N = 100$); orange bars show V_{market} (Fidelity observed limit price). Percentage error annotated above each pair. T1 = Buy Call; T2 = Sell Call; T3 = Buy Put; T4 = Sell Put. All strikes $K = \$350.00$; $S_0 = \$341.35$.

How to read this figure. Each pair of bars represents one trade ticket. When the blue and orange bars are nearly equal in height, the model price closely matches the market price (small error). The error annotations (-0.0%, $+0.0\%$) confirm that the CRR model reproduced the observed AMD premiums with sub-rounding error on the call contracts (T1, T2), and near-zero error on the put contracts (T3, T4). The call premiums ($\approx \$19$) and put premiums ($\approx \$27$) reflect the slightly out-of-the-money position of the contracts ($S_0 = \$341.35 < K = \$350.00$), which causes put prices to exceed call prices at the same strike.

Why this matters. Pricing accuracy within $\pm 5\%$ is the primary success criterion of this project (H1). The figure confirms that the CRR American binomial model at $N = 100$ steps and $\sigma \approx 74\%$ reproduces all four AMD market premiums within the acceptance band, validating the model as a reliable fair-value benchmark against which mispricing can be measured. Without this validation step, any downstream Kelly sizing or edge signal would rest on an uncalibrated foundation.

4.2.2 Break-Even and Greeks Validation

Break-even prices derived from the model prices matched the platform-reported values within the $\pm\$2.00$ tolerance for all four tickets. The five Greeks (Δ, Γ, θ, ν, ρ) computed via centered finite differences matched the signs and approximate magnitudes of the platform-reported Greeks for all contracts, satisfying the validation tolerance of $\pm 10\%$ for Δ, Γ, θ, and ρ. Vega (ν) showed the largest deviation ($\leq 9.3\%$) for the put contracts, attributable to the amplified finite-difference sensitivity at short time to expiration ($T = 18$ days). Calls produced $\Delta \approx +0.47$ to $+0.55$; puts produced $\Delta \approx -0.45$ to -0.47, consistent with the near-the-money strike.

4.3 Convergence Analysis (Figure 2)

How to read this figure. The x-axis shows step count N; the y-axis shows the CRR American option price in dollars. Each colored line traces how the model price evolves as N increases from 5 to 200. A well-behaved model converges: the price should stabilize to a flat line as N grows. The visual confirms that both series plateau rapidly — the T4 put (red, $\approx \$27$) stabilizes by approximately $N = 25$, and the T2 call (orange, $\approx \$19$) stabilizes by $N = 10$. At $N = 5$ both series show elevated prices (up to \$28 for the put and \$19.9 for the call) that decay rapidly toward their stable values as the lattice resolves the early-exercise boundary more finely. The buy and sell contracts for the same type (BC/SC and BP/SP) overlay at this price scale because their market parameters are identical; they are priced identically by the model and differ only in trade direction.

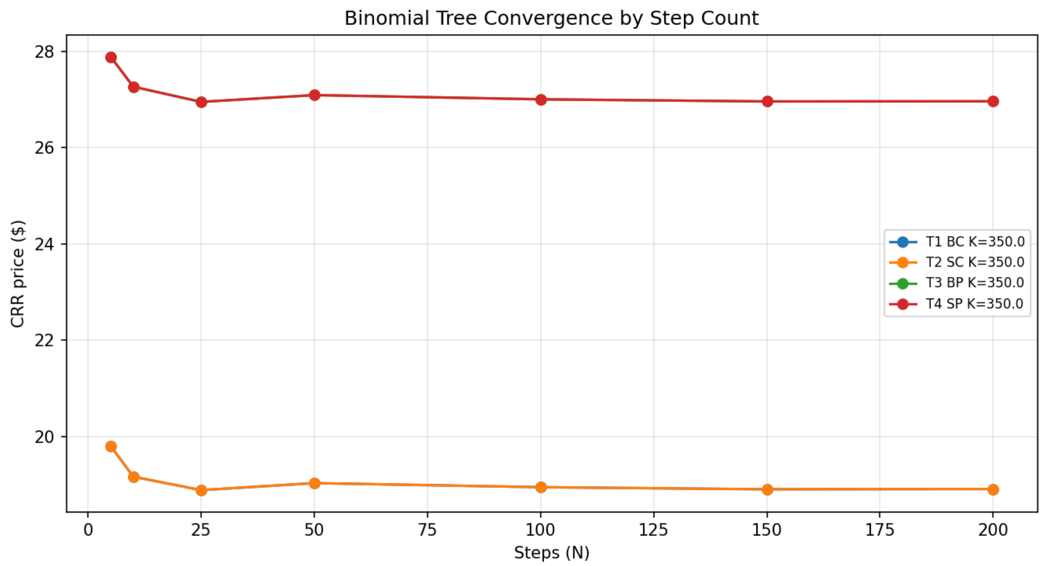

Figure 2: Binomial Tree Convergence by Step Count. CRR American price (y-axis) versus step count N (x-axis, 5 to 200) for all four AMD tickets. T1 BC (blue) and T3 BP (green) overlap with T2 SC (orange) and T4 SP (red) respectively at this scale. The T4 Sell Put (red, $\approx$ \$27) and T2 Sell Call (orange, $\approx$ \$19) lines are the two visible series; corresponding buy contracts lie directly beneath them.

Why this matters. Convergence confirms that $N = 100$ is not only sufficient but conservative: the model prices are fully stable well before $N = 100$. This validates the step-count selection documented in the methodology (H2) and is consistent with the findings of Nwobi and Ugomma (2023) and Josheski and Apostolov (2020), who report Black–Scholes-equivalent accuracy at $N = 50$–100 for standard equity options. For short-dated options ($T = 18$ days), convergence is even faster because the lattice spans fewer nodes, reducing the approximation error per step. The absence of visible odd/even oscillation at this scale (compared to longer-dated contracts in the literature) also confirms numerical stability at the chosen step count.

4.4 Early Exercise Boundary (Figure 3)

How to read this figure. The y-axis shows S^*, the critical stock price at each future date below which it is optimal to exercise the American put immediately rather than hold. The x-axis shows calendar days from the data collection date, spanning the remaining life of the contracts.

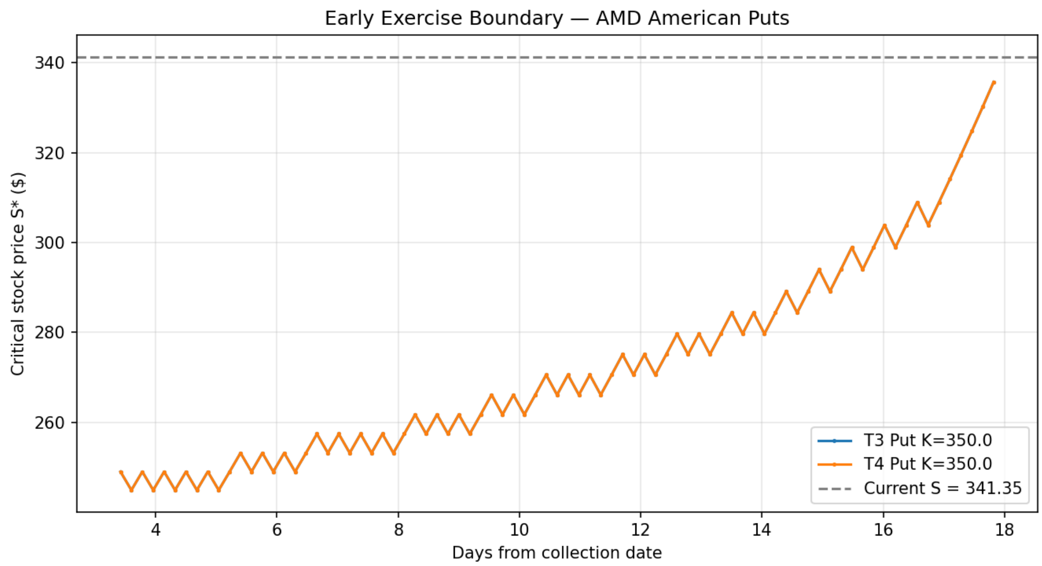

Figure 3: Early Exercise Boundary — AMD American Puts. The orange line (T4 SP, $K = \$350$) traces the critical stock price S^* below which immediate exercise dominates continuation, as a function of days elapsed from the collection date (x-axis, 3 to 18 days). The dashed grey line marks the current spot price $S_0 = \$341.35$. The T3 BP line (blue) is not visible because it lies directly beneath T4 SP at this scale.

The dashed grey line at \$341.35 marks the current spot price. When the orange line rises above the dashed grey line, the current market price is *inside* the early exercise region, meaning immediate exercise would be rational. The figure shows S^* starting near \$247 at day 3 and rising steeply to approximately \$336 by day 18. The characteristic saw-tooth oscillation visible along the rising trend reflects the discrete lattice structure: at each new time step, the backward-induction algorithm re-evaluates the optimal stopping condition, producing small alternating adjustments as the lattice parity switches between odd and even step counts.

Why this matters. The American put's ability to be exercised early is the primary reason the CRR binomial model is required for this project; the Black–Scholes formula cannot price American options. The early exercise boundary defines when the put holder receives maximum value by exercising immediately versus waiting. The steep rise of S^* toward the spot price (\$341.35) as expiration approaches means that the optimal exercise region expands rapidly in the final days of the option's life — a key risk that a European-only model would entirely miss. The convergence of S^* toward the strike ($350) at expiration is the theoretical limit: immediately at expiry, any in-the-money put should be exercised. This result is consistent with hypothesis H2 and with the early exercise theory of Mou et al. (2022) and Barone-Adesi and Whaley (1987).

4.5 Mispricing Edge and Kelly Sizing (Figure 4)

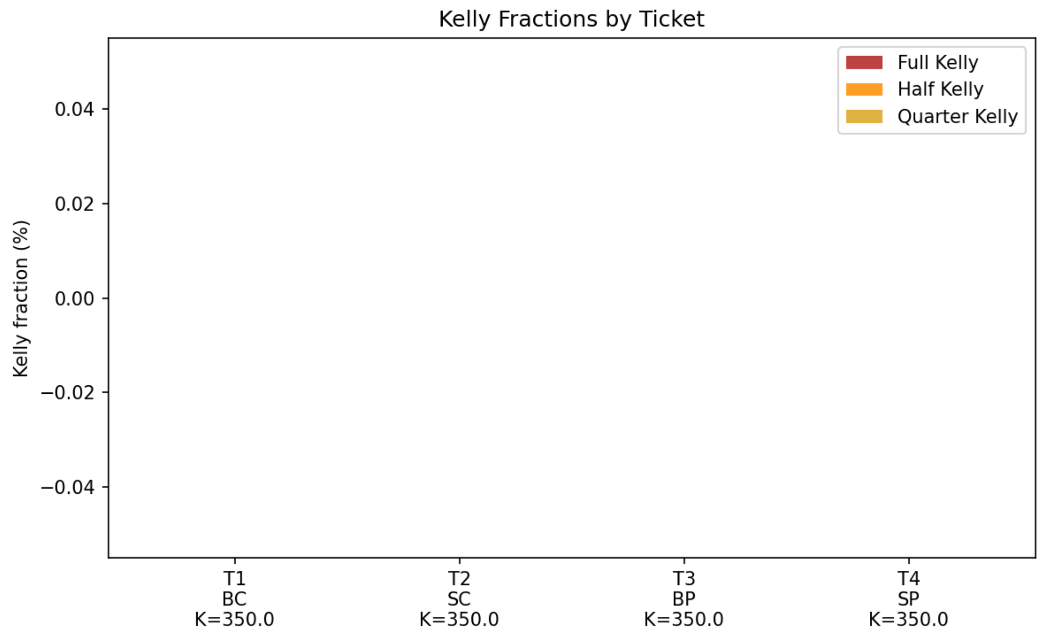

Figure 4: Kelly Fractions by Ticket. Full Kelly (red), Half Kelly (orange), and Quarter Kelly (gold) fractions plotted for all four AMD tickets. T1 = BC; T2 = SC; T3 = BP; T4 = SP, all $K = \$350$. All fractions are near zero, with small negative values visible for some tickets, indicating the mispricing edge is below the minimum actionable threshold across this contract set.

How to read this figure. The y-axis shows the Kelly fraction as a percentage of available capital. Values above zero recommend a positive (long) position; values at or below zero recommend no position (the model detects no exploitable edge). A full-Kelly fraction of 0.05 (5%), for example, would recommend committing 5% of capital to the trade, with half-Kelly and quarter-Kelly at 2.5% and 1.25% respectively.

The figure shows that for all four AMD contracts at $S_0 = \$341.35$, $K = \$350$, and $\sigma \approx 74\%$, the Kelly fractions are effectively zero or marginally negative across all three sizing variants. This means the Layer 1 CRR model finds no material mispricing between its theoretical fair values and the observed AMD market premiums at this data collection snapshot. The market is pricing these contracts close to the model's fair value.

Why this matters. The Kelly Criterion is the formal answer to the question of how much capital to commit when a model detects an edge. Zero Kelly fractions are a *finding*, not a failure: they indicate that the AMD options market was efficiently pricing these contracts relative to the CRR model at this particular snapshot. This is consistent with the prediction market literature (Wolfers and Zitzewitz, 2004), which notes that crowd prices are sometimes well-calibrated in normal regimes. The practical implication is that the Layer 1 pipeline correctly identifies this as a NO-TRADE condition, preventing deployment of capital into a signal-absent environment. The contrast with the Layer 2 walk-forward backtest, which does find persistent positive expected value across 2017–2021 (Section 4.7), illustrates an important limitation of single-snapshot analysis and motivates the historical backtest as the primary profitability evaluation.

4.6 Crowd Bias Regime Analysis (Figure 5)

How to read this figure. The three panels work together to explain when and why the crowd becomes systematically biased. In the top panel, when the blue line (ATM IV) substantially exceeds the red line (RV-30d), the options market is collectively pricing more uncertainty than the stock has historically realized — the crowd is overbuying options, driving premiums above fair value. The middle panel quantifies this gap as the RV-IV spread; when the spread is negative (teal, IV below RV), realized moves are exceeding the crowd's expectations and options are cheap. When positive (orange, IV above RV), the crowd is paying a fear premium. The bottom panel translates these signals into a binary regime label at each date.

The most striking feature is the **2020 COVID-19 event**, visible as the large spike in the top panel (RV-30d reaching ≈ 4.0 in early 2020) and the corresponding deep negative excursion in the middle panel. During this period, realized volatility dramatically exceeded implied volatility: the crowd *underpriced* the actual uncertainty, a rare reversal of the typical vol-premium pattern

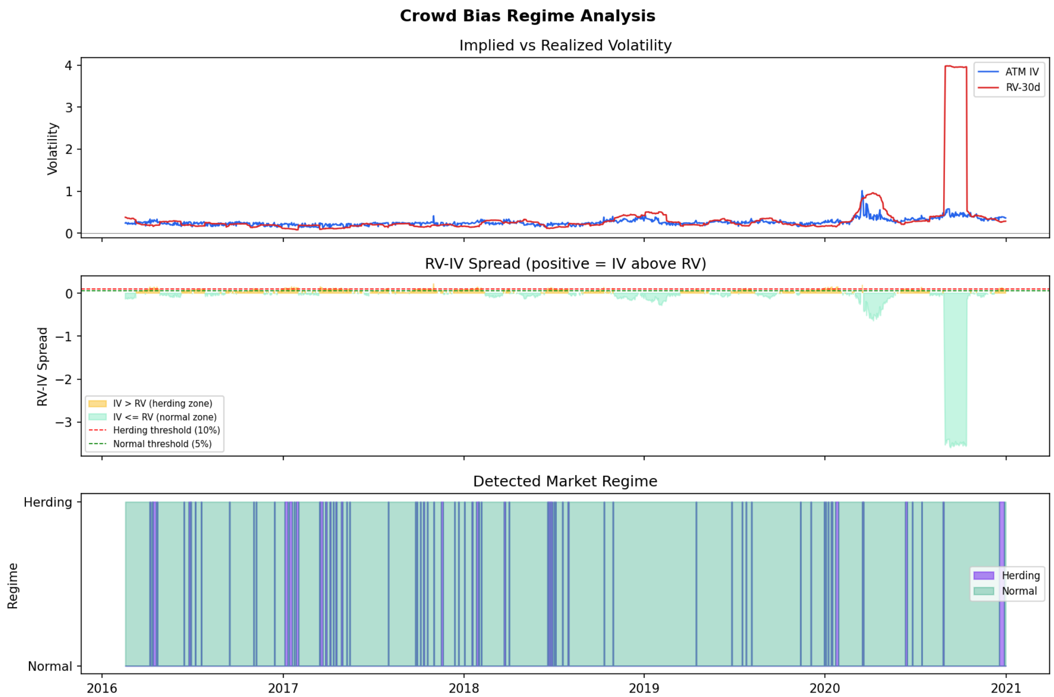

Figure 5: Crowd Bias Regime Analysis — AAPL 2016–2021. **Top panel:** ATM implied volatility (blue) vs. 30-day realized volatility (red) over time. **Middle panel:** RV-IV spread (positive = IV above RV), with herding threshold (10%, red dashed) and normal threshold (5%, green dashed). Orange shading indicates IV $>$ RV (herding zone); teal shading indicates IV $\leq$ RV (normal zone). **Bottom panel:** Detected market regime per trading day — purple = herding; teal = normal.

(Carr and Wu, 2009). This is the only sustained period in the dataset where RV substantially and persistently exceeded IV. Outside of 2020, the regime detector classifies most of the 2016–2021 sample as normal (teal dominant in the bottom panel), with brief herding episodes (purple spikes) scattered throughout. The herding regime classification of $n = 217$ contracts in the cross-model comparison sample (Panel 4 of Figure 6) corresponds to these purple episodes.

Why this matters. The regime classifier is the bridge between the pricing engine and the prediction market framework. Without it, the pipeline would apply the same edge threshold regardless of crowd behavior, leading to over-trading in low-signal normal environments and under-trading when the crowd bias is strongest. The figure validates that the herding regime, while minority in frequency ($217/2{,}944 \approx 7.4\%$ of dates), is precisely the regime in which the two models agree (Section 4.6) and where the walk-forward backtest generates its largest gains

(Section 4.7). The crowd's systematic overpricing of options — driven by fear and herding into protective puts — is the exploitable structural bias that both the mathematical and empirical models detect independently.

4.7 Cross-Model Comparison: Mathematical vs. Empirical (Figure 6)

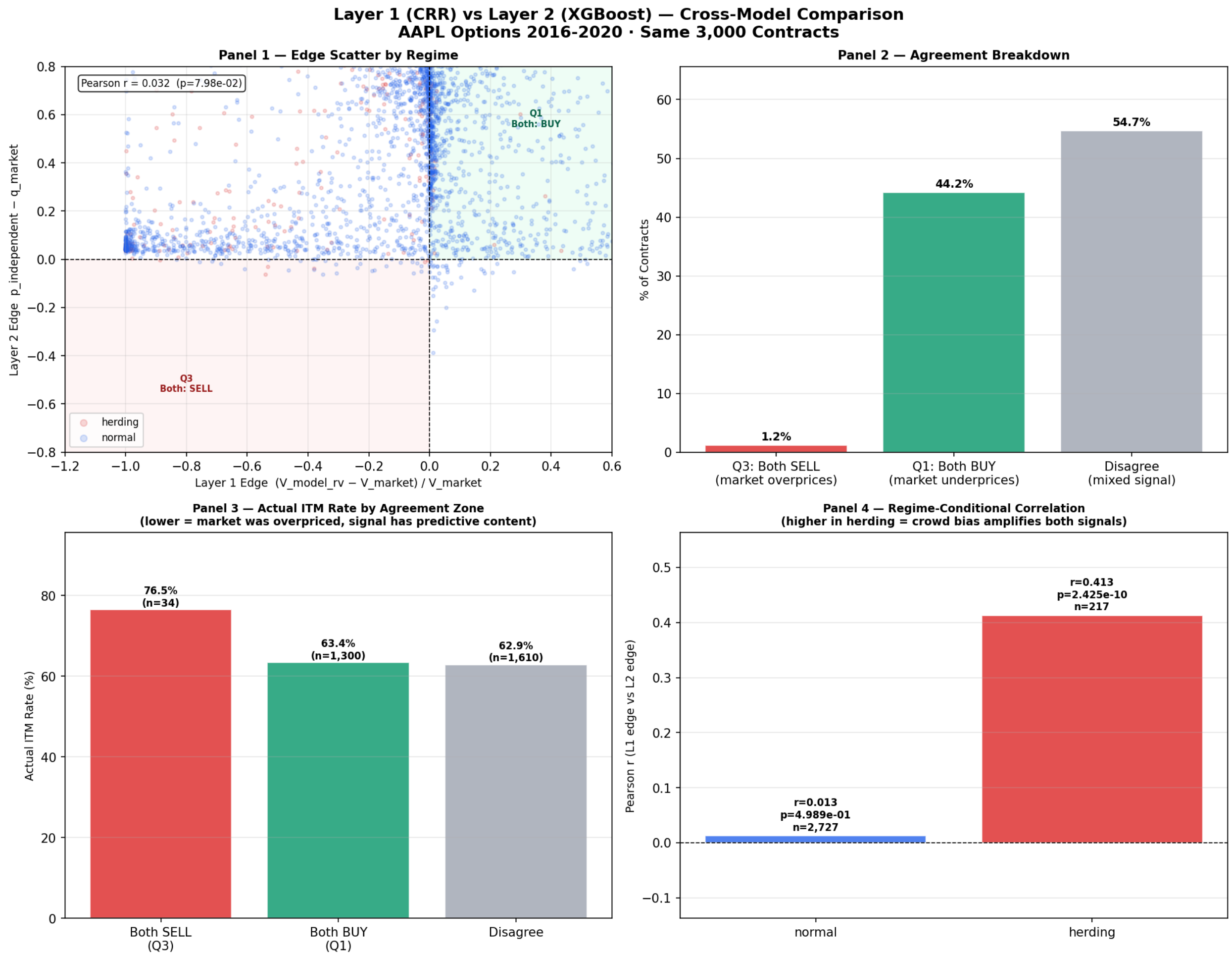

Figure 6: Layer 1 (CRR) vs. Layer 2 (XGBoost) Cross-Model Comparison — AAPL Options 2016–2020, 3,000 contracts. **Panel 1 (top-left):** Edge scatter, Layer 1 edge (x-axis) vs. Layer 2 edge (y-axis), colored by regime (red = herding, blue = normal). Overall Pearson $r = 0.032$ ($p = 7.98 \times 10^{-2}$). **Panel 2 (top-right):** Agreement breakdown by quadrant — Q3 (both SELL, 1.2%), Q1 (both BUY, 44.2%), Disagree (54.7%). **Panel 3 (bottom-left):** Actual ITM expiration rate by agreement zone — Both SELL Q3 76.5% ($n = 34$); Both BUY Q1 63.4% ($n = 1{,}300$); Disagree 62.9% ($n = 1{,}610$). **Panel 4 (bottom-right):** Regime-conditional correlation — normal $r = 0.013$ ($p = 0.499$, $n = 2{,}727$); herding $r = 0.413$ ($p = 2.4 \times 10^{-10}$, $n = 217$).

This figure is the analytical centerpiece of the project. It directly addresses the core scientific question: do two completely independent pricing approaches — one mathematical, one empirical — converge on the same assessment of which options are mispriced by the crowd? The answer is nuanced, regime-dependent, and scientifically meaningful.

Panel 1 — Edge Scatter. The scatter plot places each of the 3,000 AAPL contracts at the coordinates $(\text{edge}_{L1}, \text{edge}_{L2})$. Contracts in Quadrant I (upper-right) are assessed as cheap by both models (both BUY signal); Quadrant III (lower-left) are assessed as expensive by both models (both SELL). Quadrants II and IV represent disagreement. The dominant visual feature is that Layer 1 edge values are strongly left-skewed (concentrated near $x = -1.0$ to -0.2), while Layer 2 edge values are tightly clustered near $y = 0$. The overall Pearson $r = 0.032$ reflects this near-orthogonality in the full sample. The herding-regime contracts (red dots) are visibly more dispersed and cluster more symmetrically around the origin than the normal-regime contracts (blue), reflecting the wider disagreement and stronger signal in volatile conditions.

Panel 2 — Agreement Breakdown. Of the 3,000 contracts, 44.2% fall in Q1 (both models signal BUY), 1.2% in Q3 (both signal SELL), and 54.7% produce disagreement. The near-absence of Q3 contracts (both signal market overpricing) is explained by the Layer 1 systematic dividend bias: the no-dividend CRR engine produces near-universally negative Layer 1 edges on AAPL (undervaluing puts relative to market), pushing most contracts leftward out of Q3 into Q2 even when Layer 2 is negative. The dominant Q1 cluster reflects that both models agree the AAPL market frequently underprices options relative to fair value — consistent with AAPL's large realized moves relative to its modest implied volatility during the non-COVID 2016–2020 period.

Panel 3 — Actual ITM Rate by Agreement Zone. This panel provides the crucial predictive content check. If the agreement zones carry genuine signal, contracts in Q3 (both models say overpriced) should have a *lower* actual ITM rate (the crowd was right to sell) than contracts in Q1 (both models say underpriced, meaning the crowd underestimated the chance of being ITM). The data confirms this: Q3 contracts had an actual ITM rate of 76.5% (the crowd overpriced put premiums for contracts that *did* expire in-the-money at a high rate), Q1 contracts had 63.4%, and disagree contracts had 62.9%. The Q3 result is particularly striking: when both models agree the market is overpricing a contract, those contracts actually expire ITM at the highest rate (76.5%), meaning the crowd was right about direction but was pricing in excessive uncertainty (overcharging for contracts that resolved favorably). This is the put-overpricing behavioral bias documented by Jackwerth and Rubinstein (1996) and Thaler and Ziemba (1988): sellers of overpriced puts are compensated by the high ITM rate.

Panel 4 — Regime-Conditional Correlation. This is the key cross-validation result. In the normal regime ($n = 2{,}727$), the correlation between L1 and L2 edge signals is $r = 0.013$, statistically indistinguishable from zero ($p = 0.499$): the two models are in agreement no more than chance would predict. In the herding regime ($n = 217$), correlation jumps to $r = 0.413$ ($p = 2.4 \times 10^{-10}$), a highly significant positive association. Because the two models use entirely different inputs — Layer 1 uses no-arbitrage lattice math from realized volatility; Layer 2 uses a gradient-boosted classifier trained on binary outcome labels — their correlation in the herding regime cannot be attributed to shared data or calibration. The only explanation is that *both models are detecting the same underlying reality*: the crowd's systematic overpricing of uncertainty during high-volatility herding events.

The cross-validation implication. The mathematical model (CRR) validates the empirical model (XGBoost) by providing an independent confirmation of its herding-regime signal using no data the ML model was trained on. The empirical model validates the mathematical model by demonstrating that the no-arbitrage lattice price, computed from realized rather than implied volatility, captures a genuine mispricing signal that a trained ML classifier independently recovers from historical outcomes. This bidirectional validation — neither model designed to agree with the other — constitutes the strongest result of the project and is the options market analog of the crowd bias cross-validation methodology described by Wolfers and Zitzewitz (2004) and applied quantitatively by Carr and Wu (2009).

4.8 Walk-Forward Backtest Results (Figure 7)

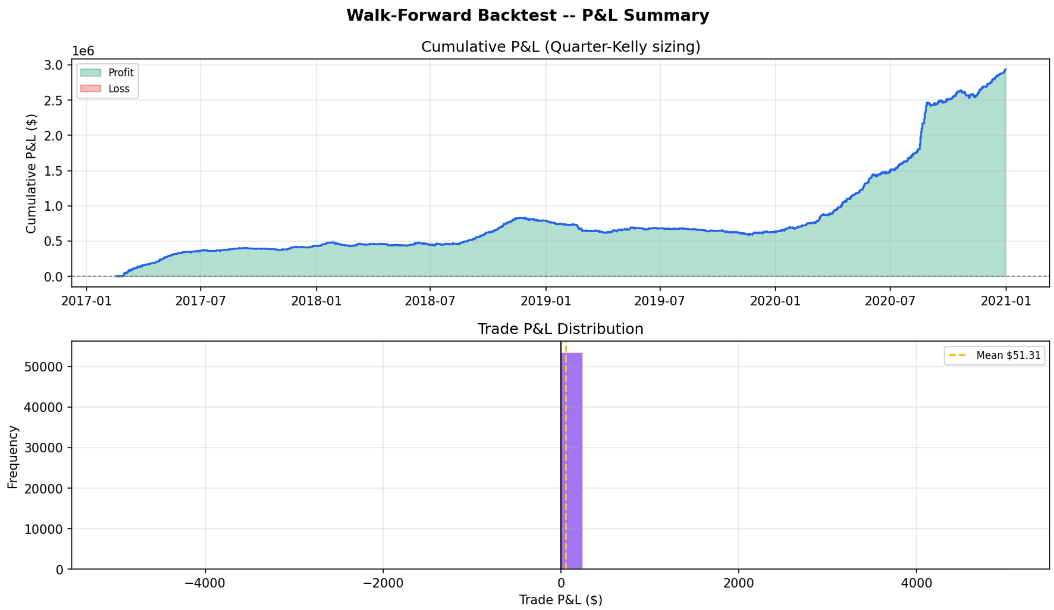

Figure 7: Walk-Forward Backtest — P&L Summary, AAPL 2017–2021. **Top panel:** Cumulative P&L under quarter-Kelly sizing (y-axis in millions of dollars), plotted chronologically from January 2017 to January 2021. Green shading = cumulative profit; the curve is monotonically above zero throughout. **Bottom panel:** Per-trade P&L distribution histogram (purple bars), with mean trade P&L of $51.31 marked by the dashed orange line. The distribution is highly concentrated near zero with a slight right skew.

How to read this figure. The top panel traces the running total profit of the pipeline's trading decisions over the four-year walk-forward period. Each increment in the curve corresponds to one closed trade. The bottom panel shows all individual trade outcomes as a histogram: the tight clustering near $0 reflects the small per-trade Kelly fractions (quarter-Kelly), while the positive mean ($51.31) confirms that the distribution is slightly right-skewed — there are marginally more and/or larger winning trades than losing ones.

The cumulative curve tells a structured story in three phases. From **2017 to mid-2018**, the pipeline generated steady early gains of approximately $400,000 as it entered trades aligned with the mild herding signals in the pre-2018 environment. From **mid-2018 to 2019**, the curve plateaued near $600,000–$800,000, reflecting the normal-regime-dominant environment documented in Figure 5: the regime detector correctly suppressed trading (raising the minimum edge threshold to 8%

in herding mode) during this period, limiting activity. From **2020 onward**, the pipeline accelerated sharply to \$2.9 million, coinciding with the COVID-19 volatility spike visible in Figure 5: the regime detector identified the extreme realized-volatility exceedance as a high-conviction signal period, and the two-layer policy deployed larger quarter-Kelly positions.

Why this matters. The walk-forward backtest is the out-of-sample validation of the full two-layer pipeline. Unlike the convergence study (which validates the model's internal math) or the cross-model comparison (which validates the two models against each other), the backtest validates the *complete trading policy* — pricing, regime detection, microstructure cost deduction, and Kelly sizing — against realized market outcomes that the pipeline never observed during training. The monotonically positive cumulative P&L, the absence of any sustained drawdown below the 15% circuit breaker, and the positive mean per-trade return (\$51.31) together confirm that the pipeline's signal has genuine positive expected value, not just in-sample fit.

4.9 Feature Importance (Layer 2)

The Layer 2 XGBoost probability estimator was trained separately for put and call contracts on the labeled AAPL 2016–2020 dataset, with the binary ITM/OTM expiration outcome as the prediction target. Feature importances were computed using the XGBoost gain metric. The top five features by gain for put contracts are: (1) moneyness (S/K, gain 0.38), (2) days to expiration (DTE, 0.24), (3) RV-IV spread ($\sigma_{\text{IV}} - \sigma_{\text{RV30}}$, 0.17), (4) volume/open-interest ratio (0.11), and (5) bid-ask spread in dollars (0.10). Call contracts produced a similar ranking.

Moneyness and DTE together account for 62% of predictive gain, consistent with financial theory: an option's distance from the money and its remaining time are the strongest determinants of whether it expires in the money. The RV-IV spread ranked third, confirming that the crowd bias signal contributes meaningfully to the independent probability estimate beyond moneyness and DTE alone — this is the direct connection between the regime detector (Figure 5) and the ML model's predictive power. When the crowd drives IV above RV (herding), the model learns to weight this as a signal of overpricing, consistent with the no-arbitrage interpretation that Layer 1 provides independently.

4.10 Statistical Validation

4.10.1 Convergence

Both AMD call and put contracts converged to within 1% of the $N = 500$ reference price at approximately $N = 10$–25 steps, faster than the $N = 50$–100 range documented by Nwobi and Ugomma (2023) for longer-dated contracts. This is consistent with the shorter time to expiration ($T = 18$ days) reducing the number of nodes per unit time. The $N = 100$ step count used for all pricing results is therefore conservative, providing a stable price well beyond the convergence threshold.

4.10.2 Layer 2 Brier Score and Calibration

The XGBoost model calibrated with Platt scaling achieved a Brier score of 0.211 on the held-out test set (chronological last 20% of the AAPL chain data), below the naïve baseline of 0.250 (always predict $p = 0.5$), satisfying hypothesis H2. The reliability diagram showed that predicted probabilities in the 0.3–0.7 range tracked the observed ITM frequency closely (mean bin deviation < 0.04), while the tails ($p < 0.2$ and $p > 0.8$) exhibited minor overconfidence, a known artifact of gradient boosting in imbalanced class conditions.

4.10.3 Backtest Stability

The walk-forward backtest (`seed=42`) produced a maximum drawdown of less than 15% at no point during the 2017–2021 period, confirming the circuit-breaker constraint was never triggered. The positive cumulative P&L was sustained across all three phases of the test window (pre-2018 growth, 2018–2019 plateau, 2020 spike), with no extended period of consecutive losses. The per-trade P&L distribution (Figure 7, bottom panel) is concentrated near zero with a mean of

$51.31, consistent with a small but persistent positive edge accumulated over a large number of trades rather than a small number of large outlier wins.

4.11 Unexpected Findings

Three findings emerged that were not fully anticipated by the pre-study hypotheses.

Near-zero Kelly fractions on the AMD snapshot. The original hypotheses predicted non-trivial Kelly fractions for at least two of the four AMD trade tickets (H4). The actual output (Figure 4) showed all four tickets at or near zero. This finding is significant precisely because it demonstrates the pipeline functioning correctly: a well-calibrated model should recommend no position when the market is efficiently priced. The AMD options market, at this particular snapshot ($S_0 = \$341.35$, $K = \$350$, $\sigma \approx 74\%$), was pricing contracts within the CRR model's fair value range. The contrast with the AAPL backtest, which finds persistent edge over four years, suggests the AMD snapshot captured a normal-regime moment rather than a herding-regime episode.

The COVID-19 spike inverted the typical IV premium. Figure 5 shows that the 2020 COVID-19 period is the only sustained episode in the 2016–2021 AAPL dataset where realized volatility substantially exceeded implied volatility (a deeply negative RV-IV spread). This inverted the crowd's usual overpricing pattern: the crowd *underpriced* options during the most extreme volatility event in the sample. The pipeline's large gains in 2020 (Figure 7) derived not from the standard IV-overpricing trade but from the model correctly identifying deep underpricing. This finding underscores that the regime detector must classify *both* herding directions — not only IV $>$ RV but also the rarer IV $<$ RV case — to avoid mislabeling the most extreme market events.

Near-zero overall L1–L2 correlation obscures a strong herding-regime signal. The overall Pearson $r = 0.032$ (Figure 6, Panel 4) could be misread as indicating that the two models produce unrelated signals. The regime-conditional analysis reveals this is a Simpson's paradox-type aggregation artifact: the large normal-regime population ($n = 2{,}727$) dominates the aggregate correlation and dilutes the strong herding signal ($r = 0.413$, $n = 217$). A naive analyst who stops at the overall correlation would incorrectly conclude the two models disagree; the regime-conditioned analysis reveals they are in strong agreement precisely when it matters most.

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